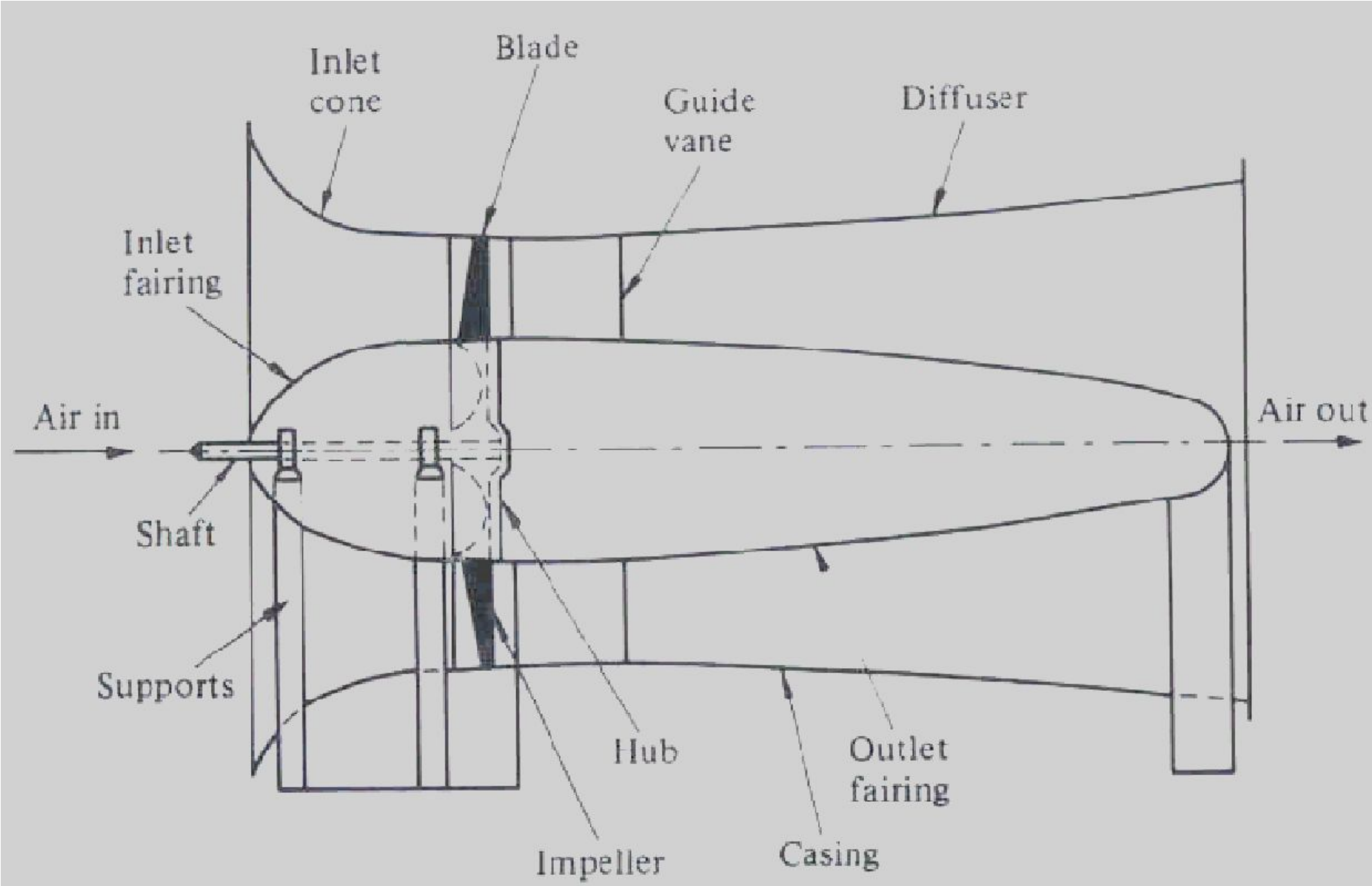
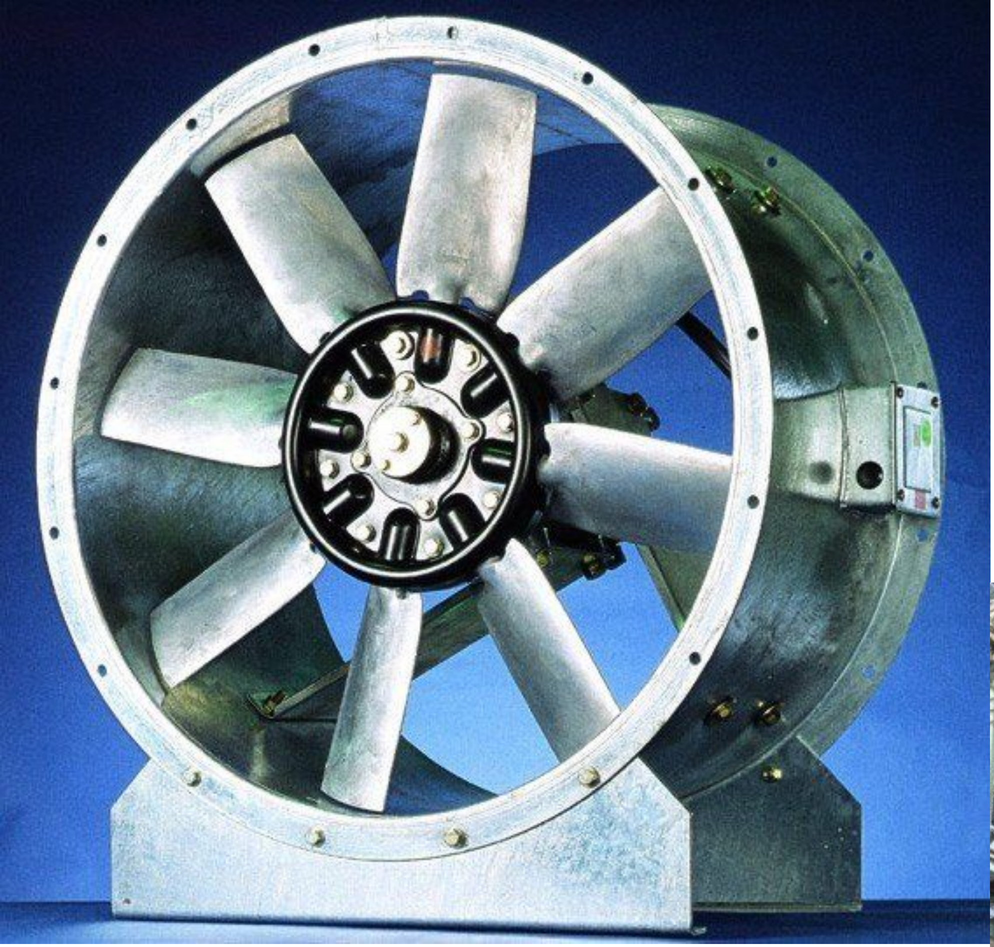
Mechanical Ventilation

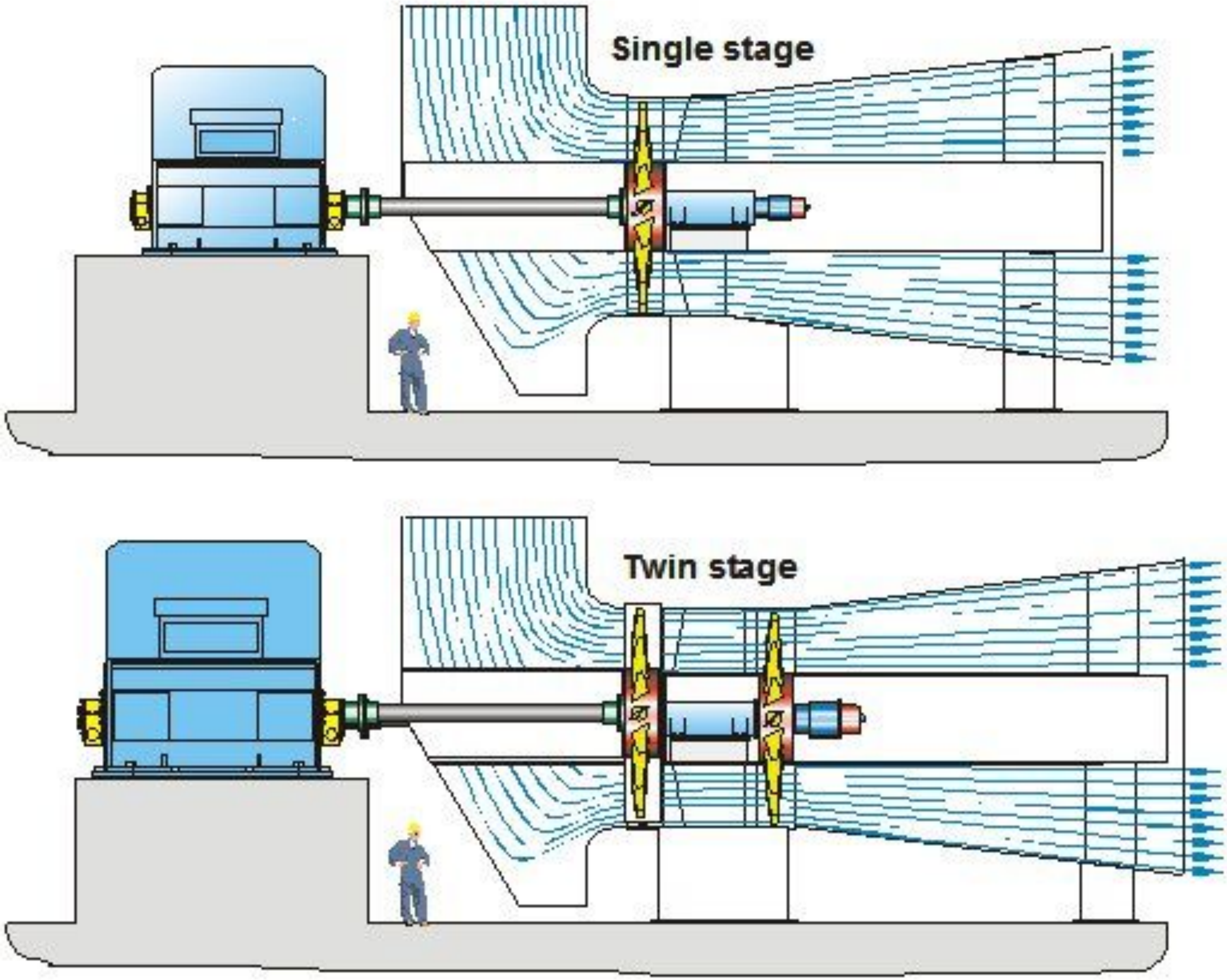
Axial flow fans

- A common example of an axial fan is a typical desk fan.
- The name "axial" comes from the fact that the air passing through the fan does not change direction and flows parallel to the fan axis.
- An axial fan is normally used when the flow requirements are high and the pressure demand is low.
- The blade/hub assembly is mounted on a shaft which rotates in a casing, and is then referred to as the rotor.
- The casing may have an open inlet, but more commonly it will have a right angled bend to allow the motor to sit outside the ductwork.
- The discharge casing gently expands to slow down the air or gas flow and convert kinetic energy into useful static pressure.



The schematic of axial flow fan

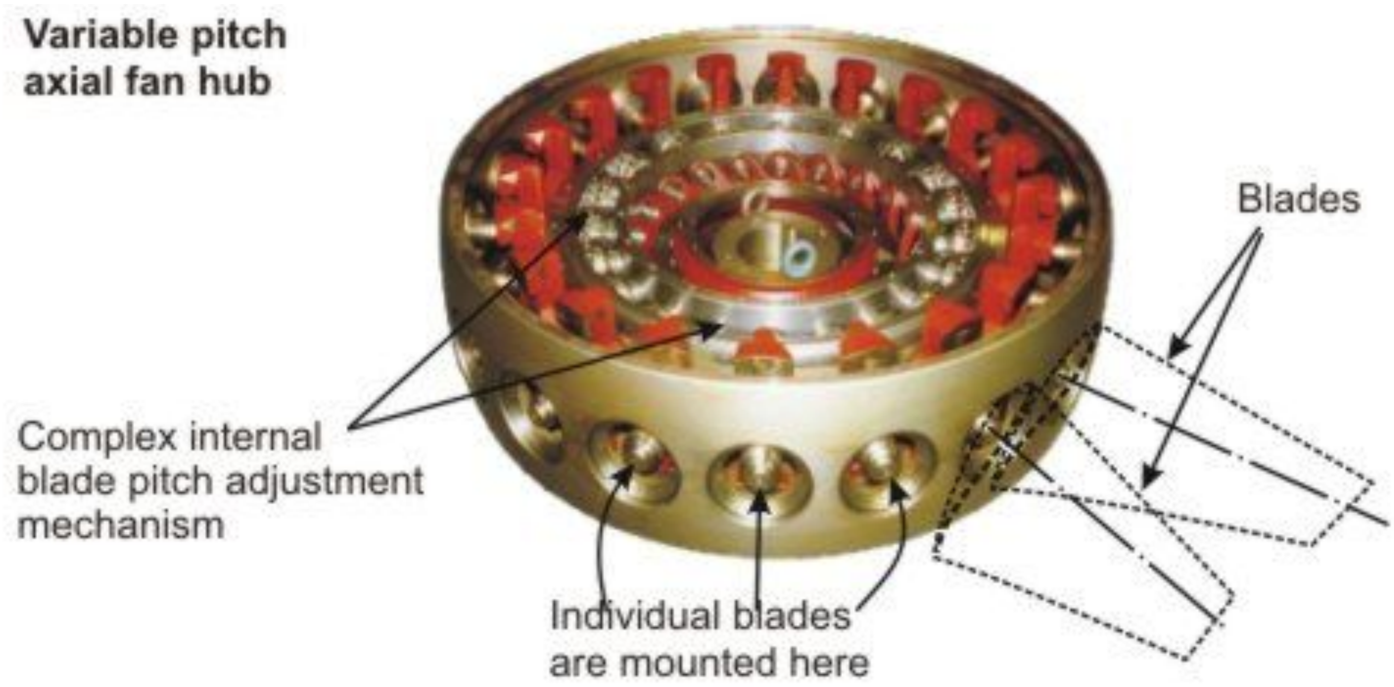




For higher pressure, two impellers can be mounted on the same shaft. **This is termed a twin-stage fan.**

- Axial fan blades are generally aerofoil in cross-section.
- The angle of the blade to the airflow, or blade pitch, can be fixed or adjustable by rotating around its longitudinal axis.
- Changing the blade angle, or pitch, is one of the major advantages of an axial fan.
- Small blade pitch angles produce lower flows while increasing the pitch produces higher flow.
- Sophisticated axial fans are capable of having the blade pitch changed when the fan is running, (like a helicopter rotor), changing the flow accordingly. These are called variable pitch (VP) axial fans.

Variable Pitch Axial Fans



- At its heart, the variable pitch (VP) axial fan has a sophisticated hub which carries the blades.
- Each blade is connected to a spindle, which is rotated by a lever.
- A servo-controlled hydraulic cylinder moves all the levers simultaneously while the fan impeller is rotating. This varies the output of the fan.

Losses and Efficiency of Fans

Various losses in fans

- The energy consumption of a fan depends to a large extent on the losses involved.

- The effect of several losses occurring in a fan give rise a lower head as well as less than 100 % flow volume and efficiency, thereby causing much energy consumption.

Therefore the study of fan losses has following three objectives:

i) To have information about the nature and magnitude of losses for knowing the way to reduce the losses.

ii) To pre-determine the head-capacity curve of a new machine from the known losses.

iii) Since, the Q – H curve of an idealized fan is a straight line, the shape of the head capacity curve of an actual fan is determined by the hydraulic losses.

Various losses that are involved in mine fans are

- The hydraulic losses,
- The mechanical losses, and
- The leakage and recirculation losses.

The hydraulic losses:

- Hydraulic losses are caused by friction between the air and the surfaces coming in contact with it.
- Also the eddy and separation losses at the impeller inlet and outlet are caused due to change in direction and magnitude of the velocity of flow.
- There is some pressure loss due to shock at inlet and outlet of impeller and guide vanes.

Air friction losses in fan occur in several ways such as

- friction between the air and the casing.
- friction between the impeller and the air, also called impeller or disc friction loss.

Friction loss in casing:

It is due to sudden change in the velocity and direction of the air current.

The general formula for such friction loss is given by

$$hf = f.\frac{L}{2D}\frac{V^2}{g}$$

Where,

f = frictional coefficient

L = length of the channel through which air flows

D = diameter of the channel section

V = velocity of air at the section with diameter D

- Since, selection of a suitable friction coefficient is a problem, several investigators combine all the friction losses in one term and the above formula in the more simplified form is as follows:

$$hf = K. \left(\frac{V^2}{2g} \right)$$

Where

K is a constant for a given type of machine and includes all lengths, areas and area ratios, and friction coefficients.

Impeller friction losses:

- This is primarily a hydraulic loss and usually called impeller friction or skin friction loss.
- The drag component causes loss due to skin friction and eddies in the wake behind the vane.
- To reduce this, proper streamlining, shaping and polishing of impeller vanes are necessary.
- The blade skin friction losses can be expressed as:

$$\mathbf{(HP)_f = k.n^3.D^5}$$

Where, k = blade skin friction coefficient

D = impeller outlet diameter

n = rpm

(HP)f = H.P. loss due to blade skin friction

The mechanical losses:

- When the fan rotates there is a frictional loss in the fan bearings, the only place where solid surfaces are in contact.
- The friction in the bearing and seals causes mechanical losses in fans.
- In the majority of cases, mechanical or bearing friction losses are very small and it is usual to assume that it amounts to about 1% of the power necessary to drive the fan.

The mechanical losses or bearing losses depend upon:

- The shaft size,
- The weight of the rotor,
- The peripheral velocity of the shaft journal,
- The coefficient of friction (μ) at the bearing [may be 0.0015 for ball bearing and 0.005 for plain type bearings], and
- Type of seals.

If, G = weight of the rotor, and

u' = the peripheral velocity of the shaft journal;

Then the work done in overcoming friction of a plain bearing is

$$N_{mech} = \mu \frac{Gu'}{75} \quad \text{HP}$$

<u>The leakage and recirculation losses</u>:

- This type of loss occurs due to leakage and recirculation of air within the fan.

- Leakage and recirculation can take place

  - at the edges of the blades where these run in a fixed casing, or
  - in the form of eddies between adjacent blades.

- In radial-flow fan there is a natural tendency for air to leak back from the periphery towards the centre and so there is recirculation.

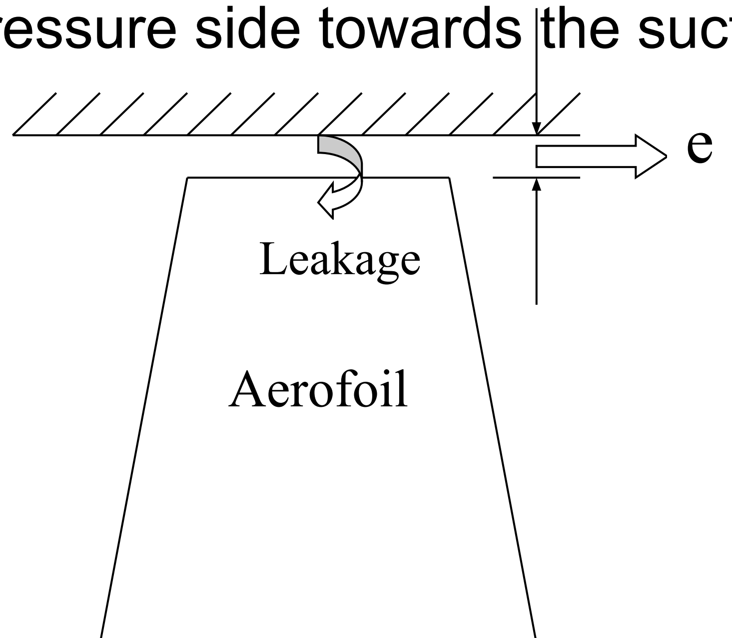
- In case of multi stage fans the loss occurs between the two adjacent stages.

Tip-clearance Loss:

- This kind of loss arises due to the clearance between a moving blade and the casing.

- The clearance in between the blade tips and the outer casing in case of axial flow fans permit recirculation of air from the outlet to the inlet side.

- On account of the static pressure difference in the operating fan, the flow leaks from the pressure side towards the suction side



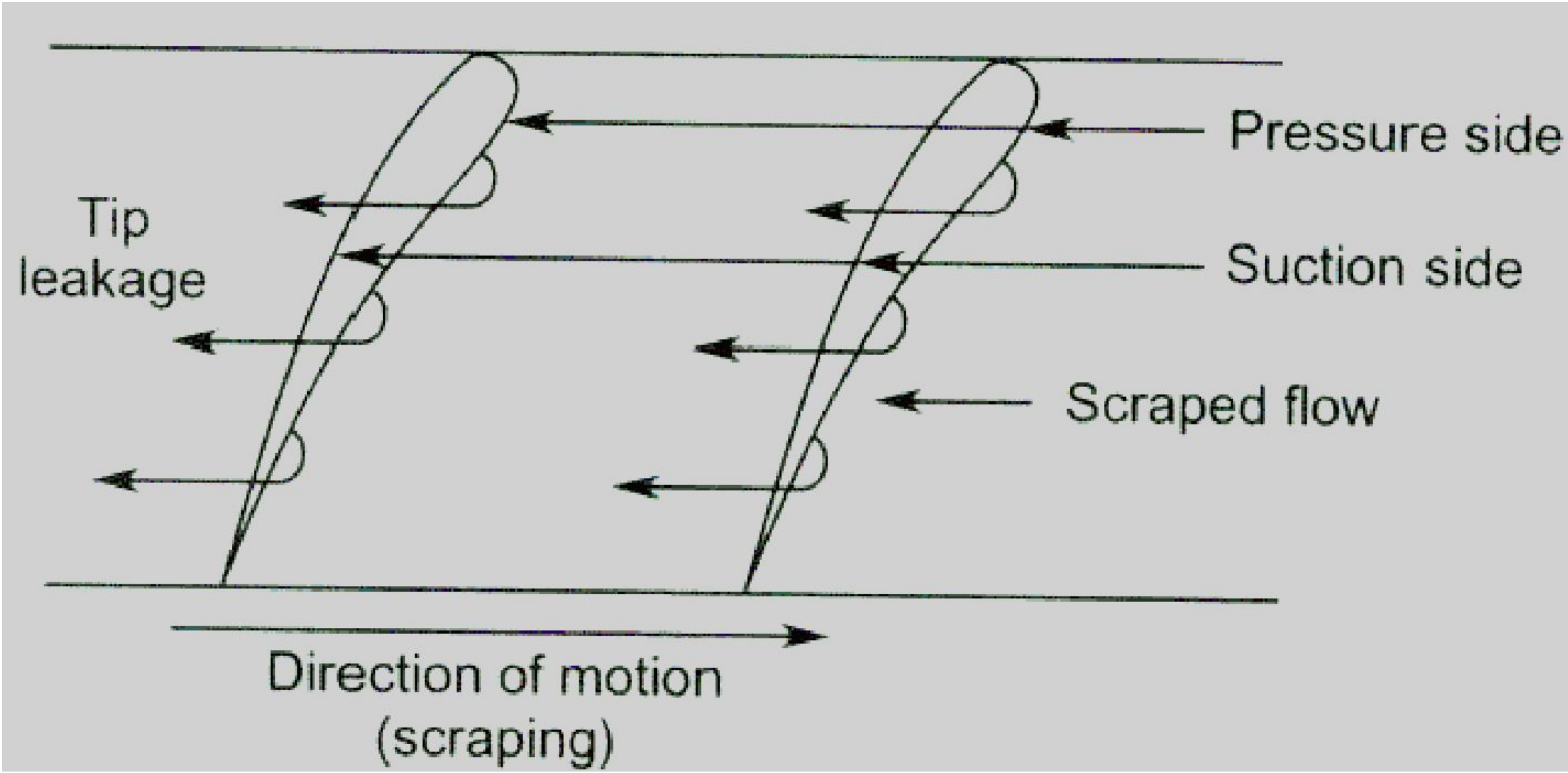


Figure. Leakage of air across the aerofoil tip from the pressure surface to the suction surface

- The efficiency loss varies linearly with tip clearance and can be expressed as

$$\eta = 2 \{(e/b) - 0.01\}$$

Where,

e = tip clearance and

b = blade height.

- The greater the impeller-to-casing distance, the more is the energy consumption.

- It is recommended that the average tip clearance should be 0.2 % to 0.5 % of the fan diameter for maximizing fan performance.

- The maximum clearance under no circumstances should exceed 0.3% of the impeller diameter.

Table: Approximate values of various losses in axial flow fan
(By MacFarlane)

Loss Components	**% Loss**
A. Drag loss (Wheel, guide vanes etc.)	9.1
B. Swirl component	6.16
C. Axial component	12.0
D. Velocity head at outlet	3.65
E. Bearing losses	1.00
Total loses not recoverable (A + D + E)	13.75
Total losses recoverable 70% of B + 80% of C	13.91
Total loss (nett)	18.00

Fan static efficiency, η_s = (100 - 18) = 82.00
Fan total efficiency, η_t = (82 + 3.65) = 85.65

Efficiency of fans

- Efficiency is a means for analyzing fan performance.

- Like any other machine, the efficiency of a mine fan is the ratio of useful power output to power input and expressed in percentage.

- According to the types of losses involved, it is advantageous to categorized fan efficiencies as follows:

1. Hydraulic efficiency:
2. Volumetric efficiency:
3. Mechanical efficiency:
4. Static efficiency:
5. Total efficiency:
6. Overall efficiency:

<u>1. Hydraulic efficiency</u>:

- It is defined as the ratio of the actual pressure head developed by the fan to the theoretical input head or Euler's head.
- Also it is referred as ***manometric efficiency*** in the literature.
- The hydraulic efficiency of fan, $\eta_{hyd} = \dfrac{\Delta P}{\Delta P + \Delta P_1 + \Delta P_{imp} + \Delta P_{diff} + \Delta P_{shock}}$

Where,

ΔP = Actual pressure head developed at the fan

ΔP_1 = Pressure loss due to bend at entry to the impeller

ΔP_{imp} = Pressure loss due to friction in impeller

ΔP_{diff} = Pressure loss in conversion of kinetic energy to pressure energy in diffuser

ΔP_{shock} = Shock pressure loss at impeller and guide vane entries

- The hydraulic efficiency of fan, η_{hyd} is strongly affected by
  - The flow passage configuration of the fan.
  - The finish of its inner surface and blades.
  - The viscosity of the liquid.

2. Volumetric efficiency:

- The flow through the impeller is greater than the measured discharged quantity by the amount of leakage.

- Volumetric efficiency gives the ratio of the outlet to the inlet quantities of air and it materially affected by the amount of tip clearance.

- With respect to the clearance loss, the volumetric efficiency η_{vol},

$$\eta_{vol} = \frac{Q}{Q + \Delta Q} = \frac{1}{1 + \frac{\Delta Q}{Q}}$$

Where,

Q = Actual quantity delivered

ΔQ = Clearance losses

- Thus, high η_{vol} can be obtained by reducing the tip clearance.

3. Mechanical efficiency: The purely mechanical losses, N_m are made up of bearing and transmission losses, i.e. gear, belt and V-belt losses.

- Mechanical efficiency can be calculated in two ways such as:

Method I: In this method N_m is to be subtracted from the total input power, i.e. brake horse power. Thus,

$$\text{Mechanical efficiency, } \eta_{mech} = \frac{\text{brake horse power – mechanical losses}}{\text{brake horse power}}$$

$$= \frac{N - N_m}{N}$$

Where, N = brake horse power (bhp)

Method II: In this method N_m is added with the impeller shaft output. Then

$$\eta_{mech} = \frac{N_W}{N_W + N_m}$$

Where,

N_W = impeller shaft output

N_m = mechanical losses (bearing and transmission losses)

4. Static efficiency (η_s):

- This efficiency is measured by calculating the air power on the basis of static head.

η_s = air power based on fan static head / input mechanical power or bhp.

5. Total efficiency: η_t = air power based on fan total head / input mechanical power or bhp.

- Where fan total head = fan static head + fan velocity head

6. Overall efficiency (η_o): The overall efficiency of fan indicates the combined efficiency of the drive motor, transmission and fan.

η_o = air power based on fan total head/ electrical input power × 100%

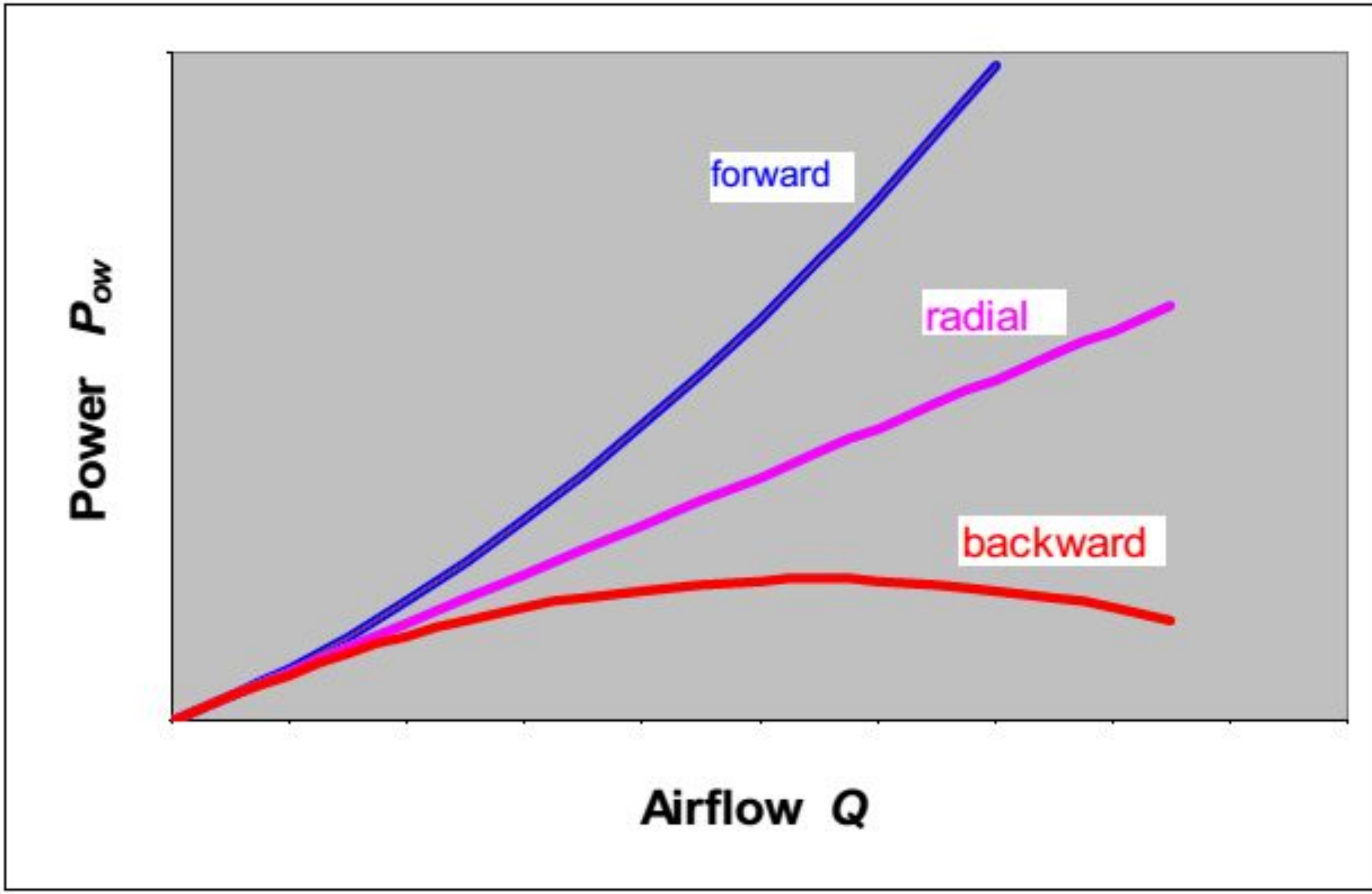
FAN CHARACTRISTIC CURVES

- The performance of a fan under various conditions can't be expressed by simple mathematical equations, but is most conveniently expressed by a **series of curves known as characteristic curves.**

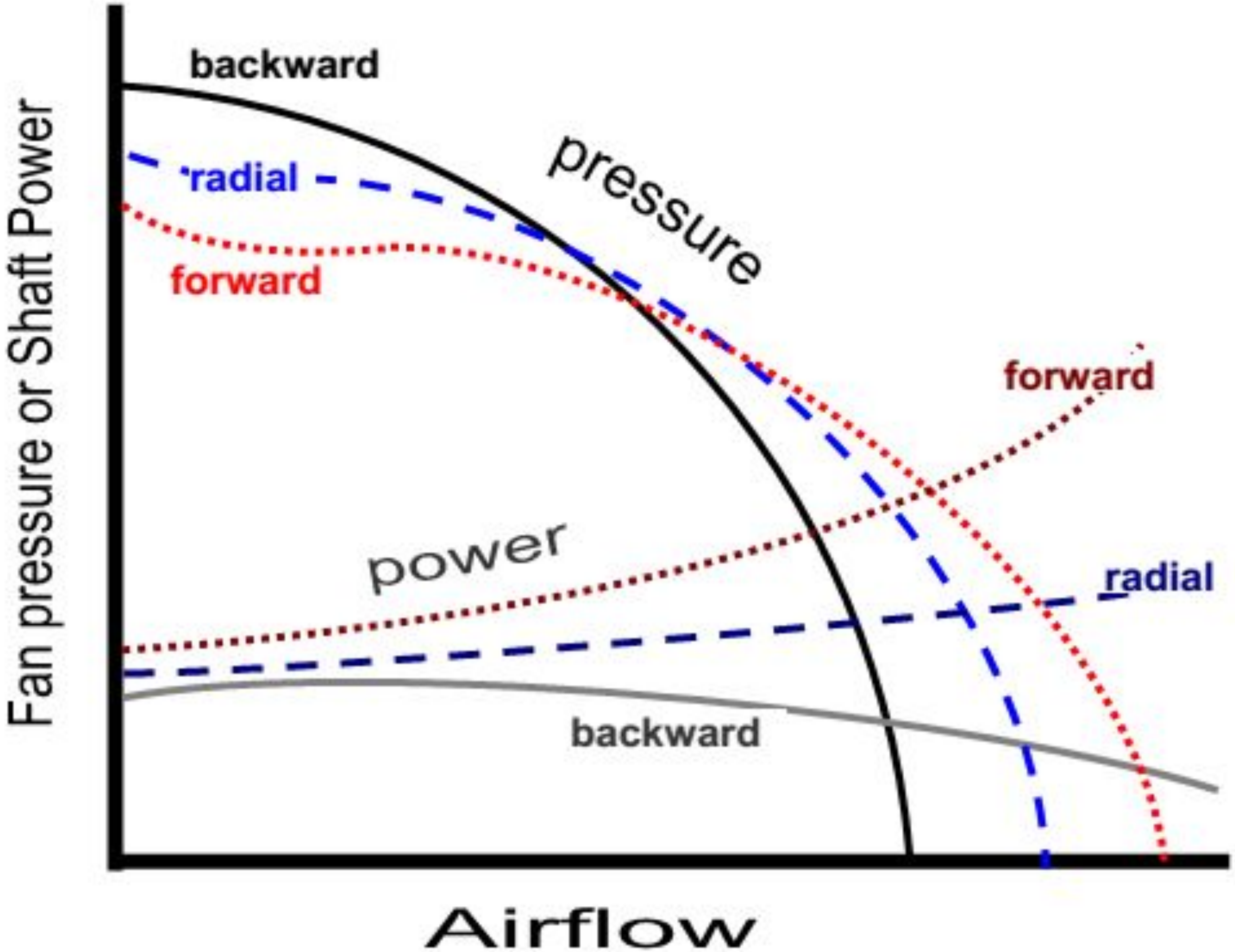
- **A fan characteristic curve is a curve which shows how the magnitude of one quantity varies with changes in some other related quantity.**

- In case of a fan for every speed, a set of curves can be drawn to show variation of fan-drift pressure, B.H.P. of the prime-mover and mechanical efficiency of the fan with changes in volume of air.

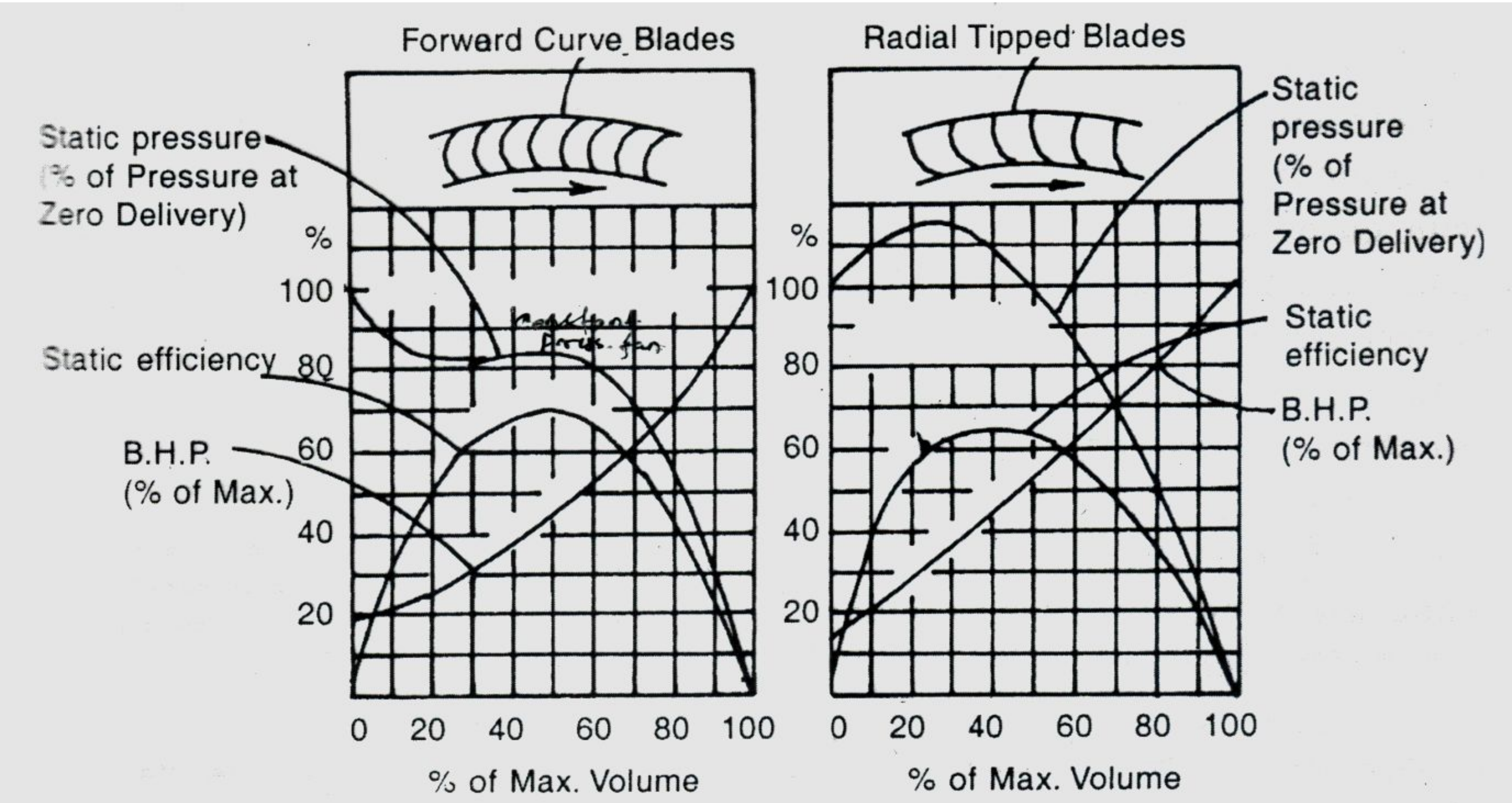
- These curves are indicative of the performance or the characteristic of the fan and hence, are referred as the 'characteristic curves' of that particular fan.



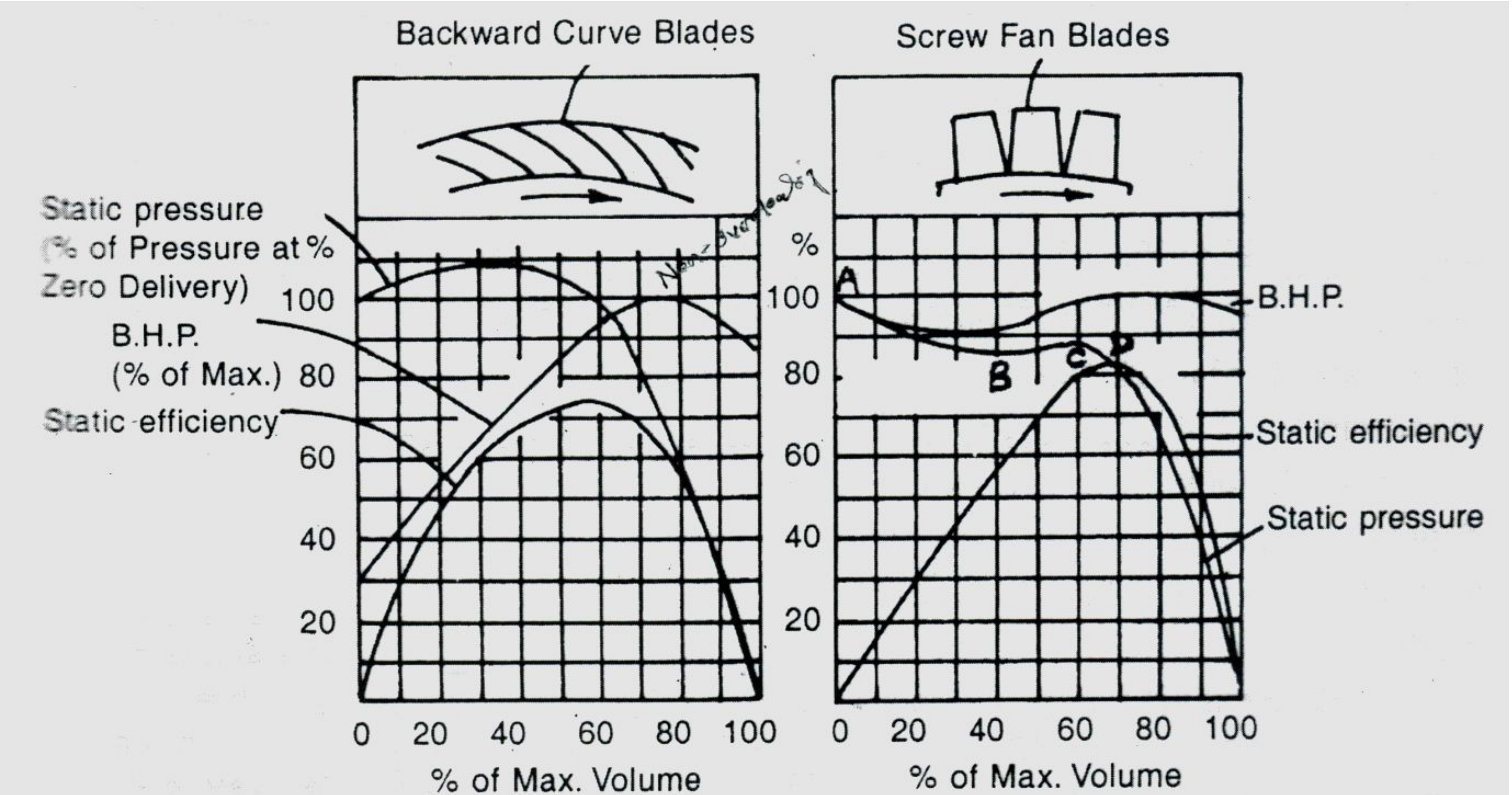
Theoretical power-volume characteristics



Actual pressure and shaft power characteristics for centrifugal fans



Typical characteristic curves of mine fans



Typical characteristic curves of mine fans

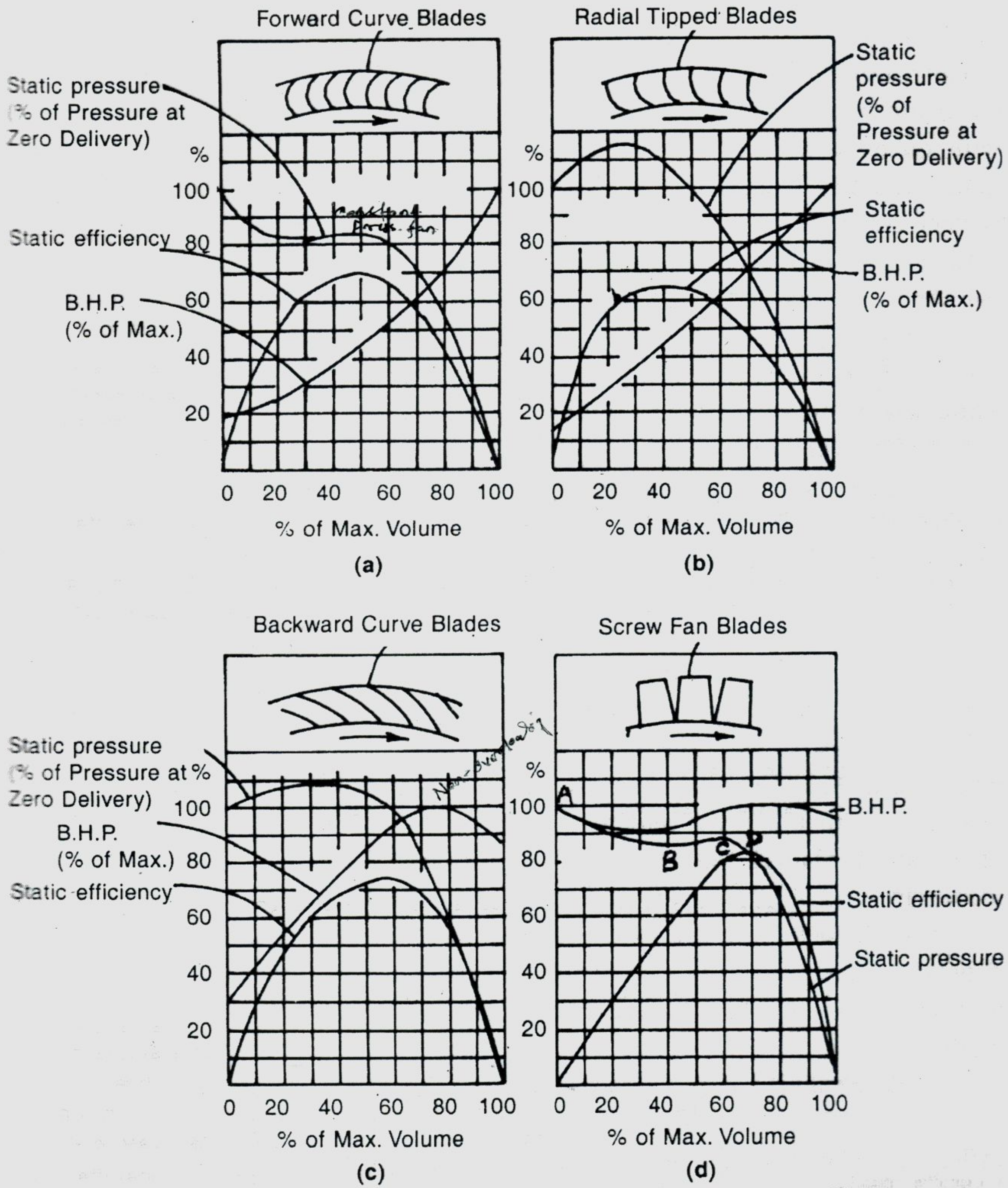


Fig. 7.14 Typical Characteristic Curves for four Types of Fans[2]

Combination of fans

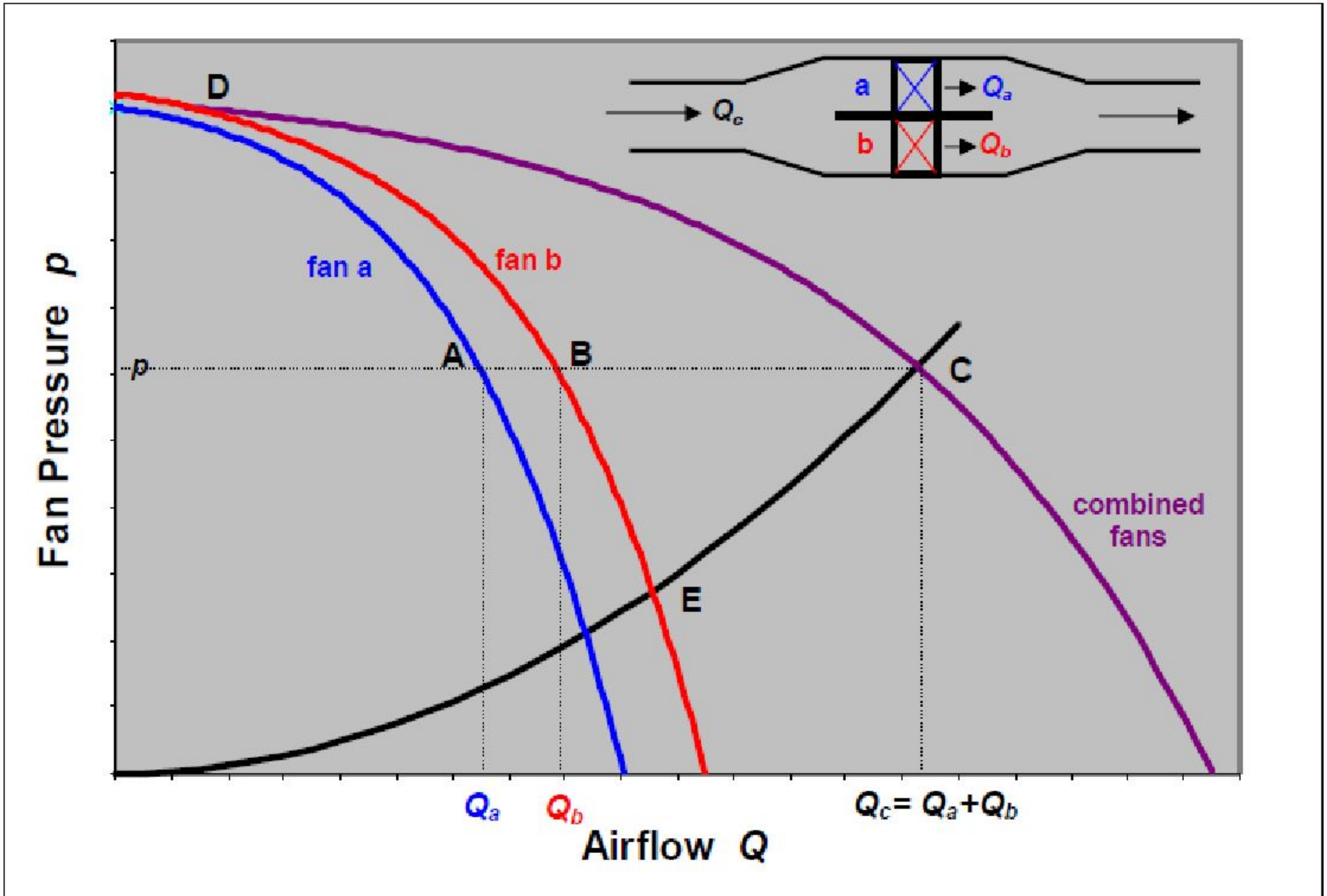


Figure 10.14 Characteristic curves for two fans connected in parallel.

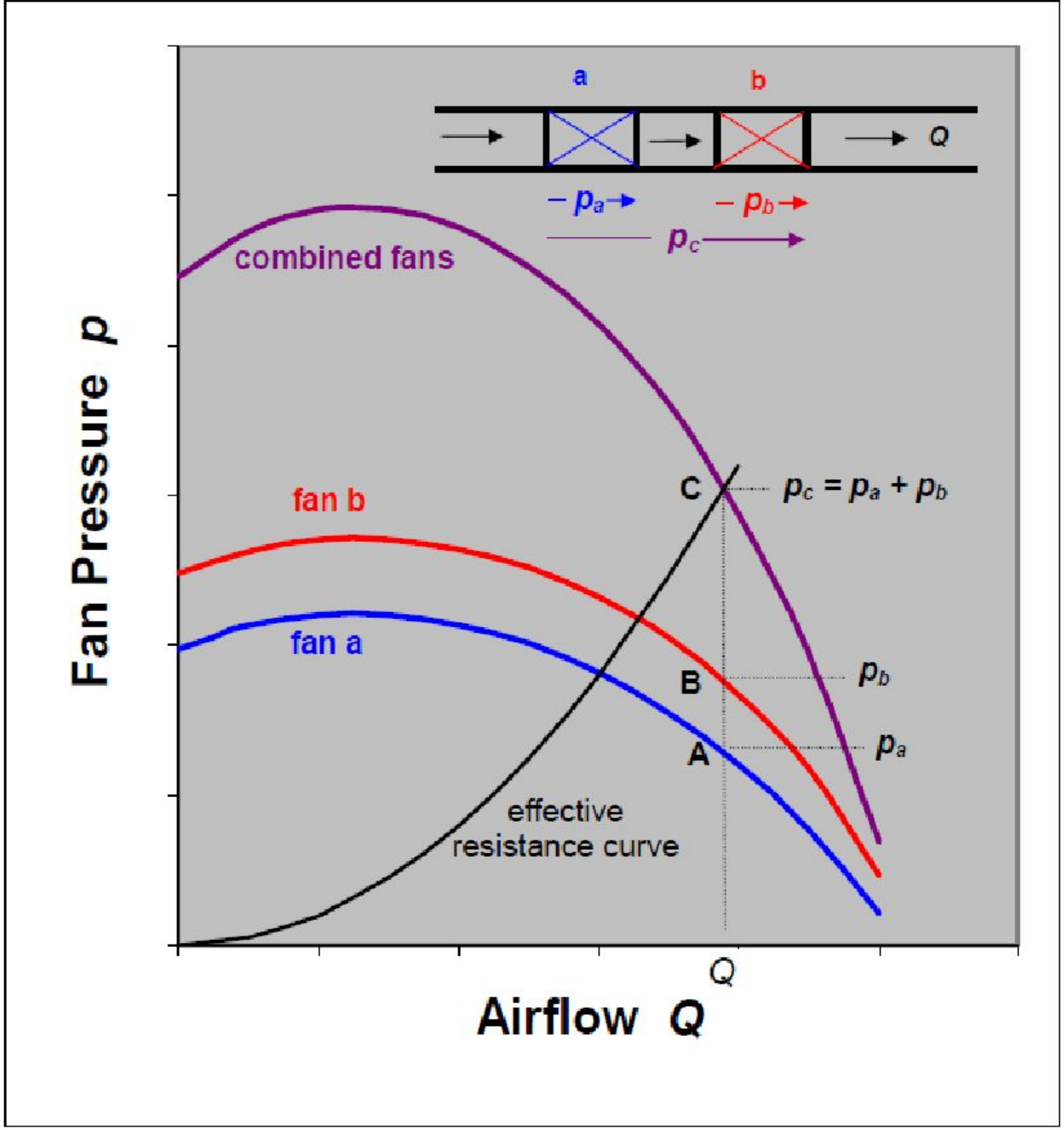
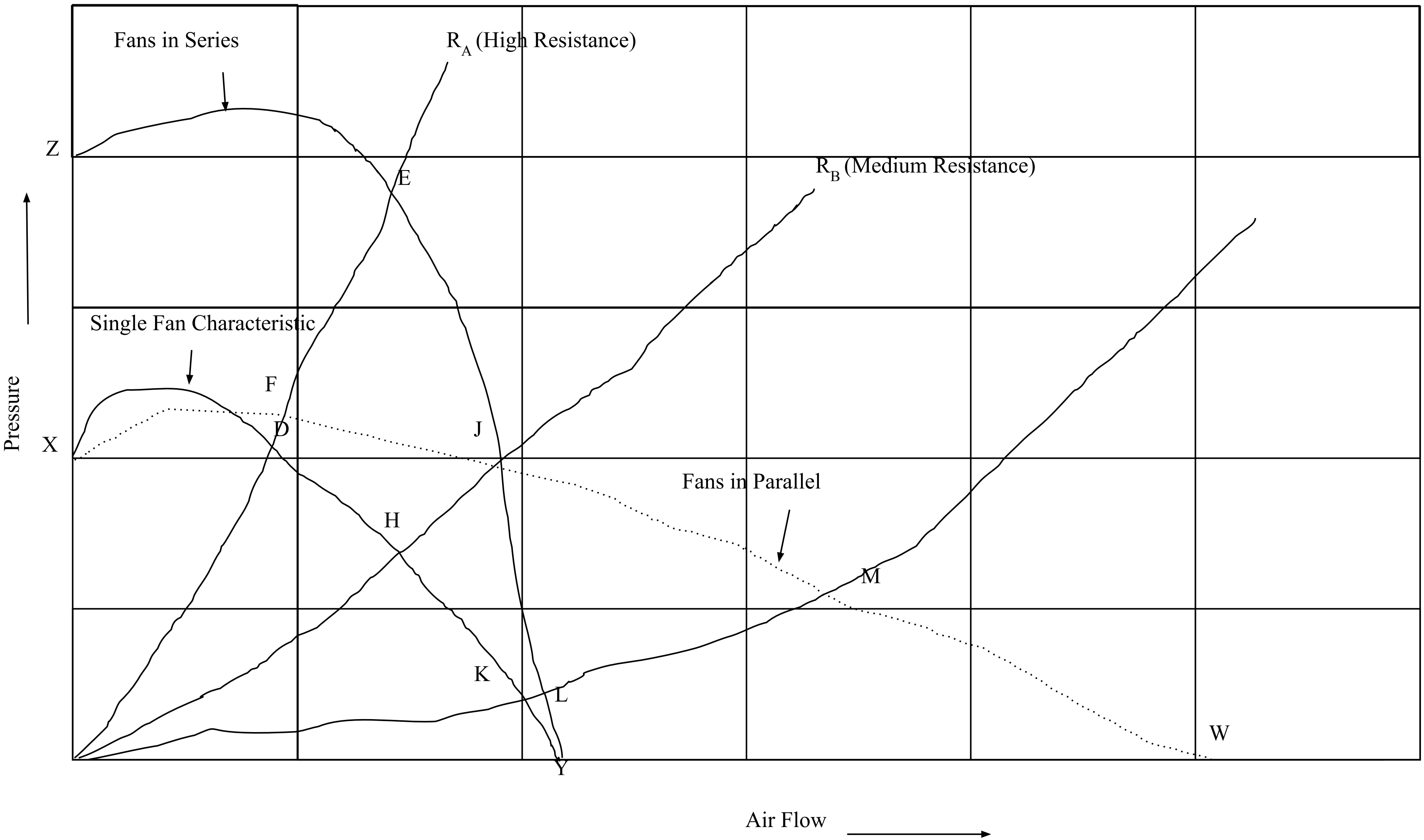


Figure 10.13 Characteristic curves for two fans connected in series.



Fans in Series
R_A (High Resistance)
R_B (Medium Resistance)
Single Fan Characteristic
Fans in Parallel
Pressure
Air Flow
Z
X
Y
W
E
F
D
J
H
K
L
M

Conclusions

- In a mine of high resistance, the series arrangement gives a considerable increase in quantity of air flowing, while the parallel one gives a negligible increase or a decrease depending on the shape of the fan characteristic curve in some cases.

- In a mine of medium resistance, both the series and parallel give the same result.

- In a mine of low resistance, where the single fan is operating at a very low efficiency on an unsuitable part of its characteristic curve. At this mine, the series arrangement gives a negligible increase in quantity, while the parallel one gives a significant increase in quantity. Thus, the parallel arrangement should be preferred in a mine of low resistance.

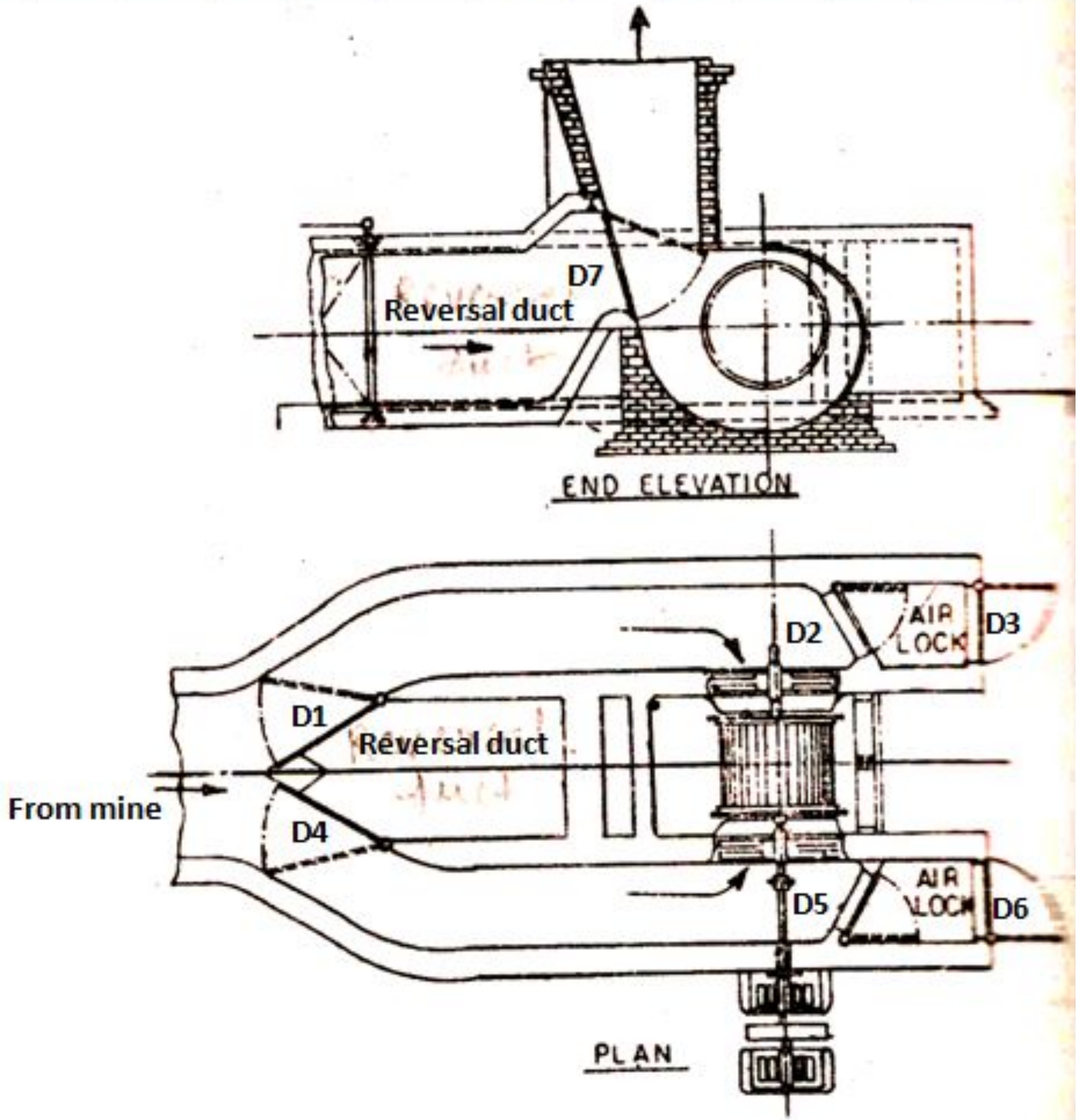


Fig. Reversal arrangement for a double inlet C.F. fan

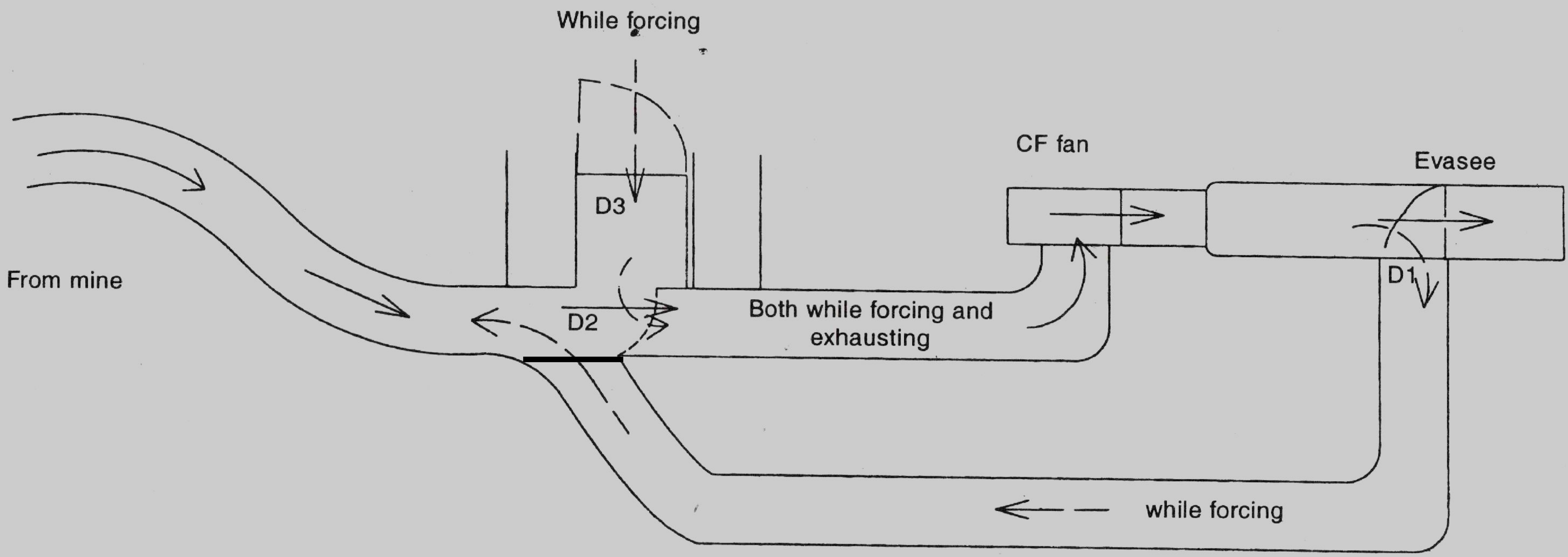


Fig. 7.28 Reversal arrangement for a single inlet C.F. fan

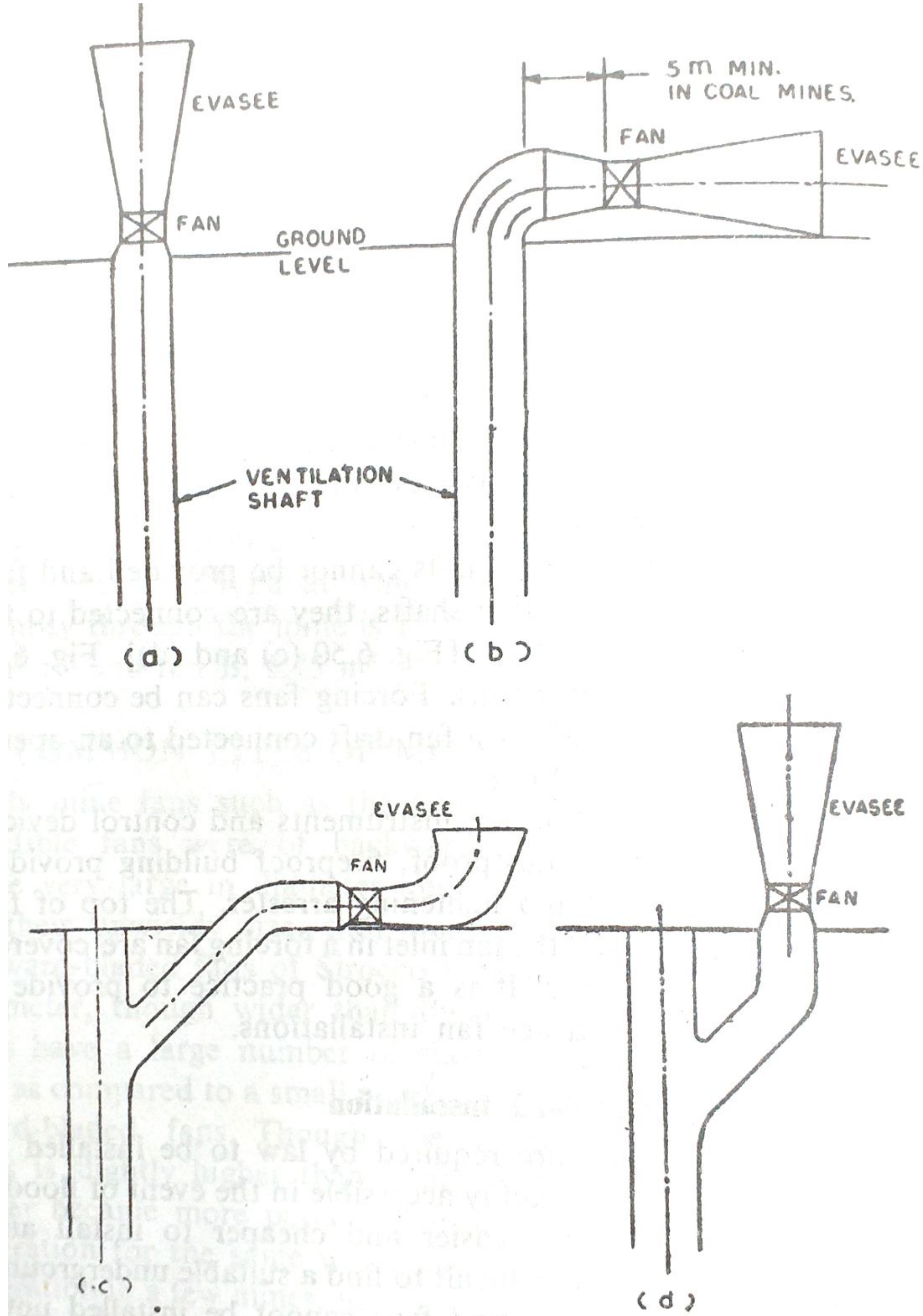


Fig. 5.50 Modes of installation of fans on shaft-tops.

Fan Laws or Affinity laws of fan.

Design or selection of a fan for a given set of conditions is based on affinity laws which gives relationship between the head, capacity, speed and size of the fan.

- When the speed (N) of a given fan is changed

1. its Capacity (Q) varies directly as the speed, ($Q \propto N$)

   or $\frac{Q_1}{Q_2} = \frac{N_1}{N_2}$, i.e. flow varies linearly with speed.

2. Head (H) varies as the square of the speed, ($\text{Head (H)} \propto N^2$)

   $$\frac{H_1}{H_2} = \left(\frac{N_1}{N_2}\right)^2$$

3. Power varies as the cube of the speed ($B.H.P \propto N^3$)

   or $\frac{P_1}{P_2} = \left(\frac{N_1}{N_2}\right)^3$

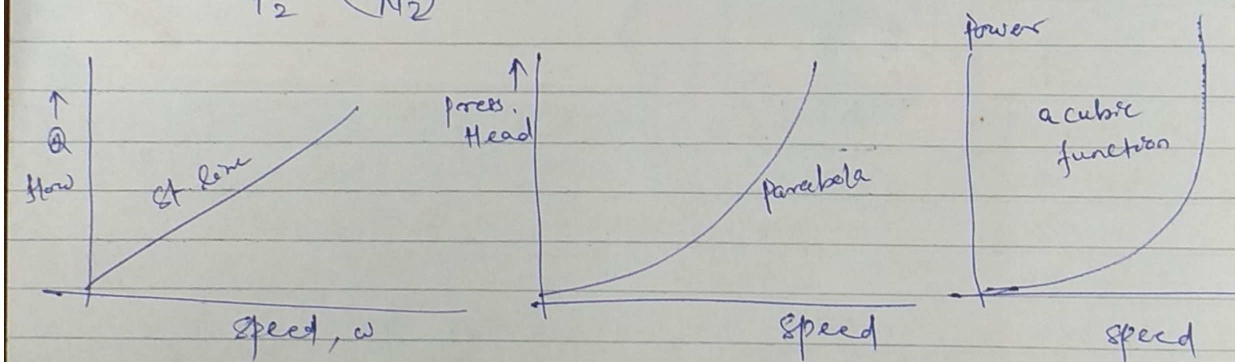


Fig: Pictorial representation of fan laws.

SELECTION OF MINE FANS

- A fan has to be selected basically to meet the pressure and quantity demands of the mine which has to be carefully estimated not only for the present but also for the future covering the life of the fan.

- The following are the major factors that have to be considered for selecting a mine fan:

1. Quantity required at various periods during the life of the mine

- The pressure-quantity duty of a fan is selected on the basis of the ventilation requirements of the mine.

- A fan should be so selected that it operates with as high an efficiency as possible during the major operational life of the mine.

- A high efficiency is essential not only from the point of view of minimizing running cost but also for keeping down noise.

2. Pressure required for circulating these quantities (i.e. the range of variation of the mine characteristics)

- The variation in mine resistance should be small so that it does not change the efficiency of the fan to a large extent.

- In case of large variation in mine resistance, other means of varying the duty of fan such as varying speed, adjusting blade pitch etc. have to be adopted.

- Where large variations in the mine resistance are likely to occur, variable-pitch axial flow fans are more flexible and should be referred.

- When the mine characteristics varies over a wide range over the life of the mine so that the operation of a single fan over its whole life becomes inefficient and uneconomical, multiple installation of fans may be considered.

- Comparatively, an air screw fan is more suited for low water gauge and high volume but a centrifugal fan, more suited for high pressure and less volume.

- Air screw fan, however, can be conveniently used for developing high water gauge by installation of stages.

3. Altitude or barometric pressure at the fan site and the temperature of air to be handled by the fan (primarily for obtaining the air density)

4. Whether reversal of airflow required

- Axial flow fans are superior to centrifugal fans since they do not need an elaborate arrangement for air reversal.

- The reversal of air current with axial flow fans can be affected by simply reversing the stator connections of the driving motor which changes the direction of rotation of the motor and hence of the fan.

- The fan when running in the reverse direction, generates only 65% of the normal volume at a lower efficiency (with a power consumption of 75% of the normal), but the reduced quantity is usually adequate to meet the exigencies of the situation requiring a reversal of the air current.

- On the other hand, air reversal with centrifugal fans requires elaborate arrangements of doors and ducts in a suitable housing involving a large capital investment.

5. Type and speed of motive power available, if any

6. Degree of permissible sound emission

- Noise in fans is a function of rotational speed.
- Since axial flow fans usually high speed, they produce more noise.
- A fan produces least noise when operating at the maximum efficiency.

7. Initial and running cost

- Axial flow fans have a higher initial cost owing to the complicated design of the impeller and the better material of construction.
- But they require a less costly motor (since high speed motors of similar power are less costly than low-speed ones) and a less costly housing.

8. Size of the fan, nature of housing and space required

- Axial flow fans can have factory made steel housing or can be installed in concrete housing while centrifugal fans are usually installed in brick or concrete housing.

- Moreover, axial flow fans weigh about 40-60% less than centrifugal fans of similar capacity thus requiring less costly foundation.

- When all these have been taken into consideration, axial flow fans cost as much or even less than modern radial-flow fans of cast aerofoil impellers for small heads.

- But for large heads where multi stage axial flow fans become necessary, centrifugal fans are cheaper in the initial cost.

- Air screw fans require less installation space compared to centrifugal fans.

9. Nature of the air to be handled, viz. humidity, dustiness, etc.

- The greater streamlining of flow through the impeller of an axial flow fan causes less accumulation of dirt on the blades than in centrifugal fans.

- This coupled with the better material of construction of axial flow fan vanes, make them less liable to corrosion.

10. Requirement of maintenance

- As regards maintenance, low-speed centrifugal fans are superior to high-speed axial-flow ones.

- In the former, the bearings are accessible and the drive, always isolated from the air stream whereas only large size axial flow fans have isolated drives.

- It must be remembered that an isolated drive is much preferred from the point of view of maintenance.

Booster Fan

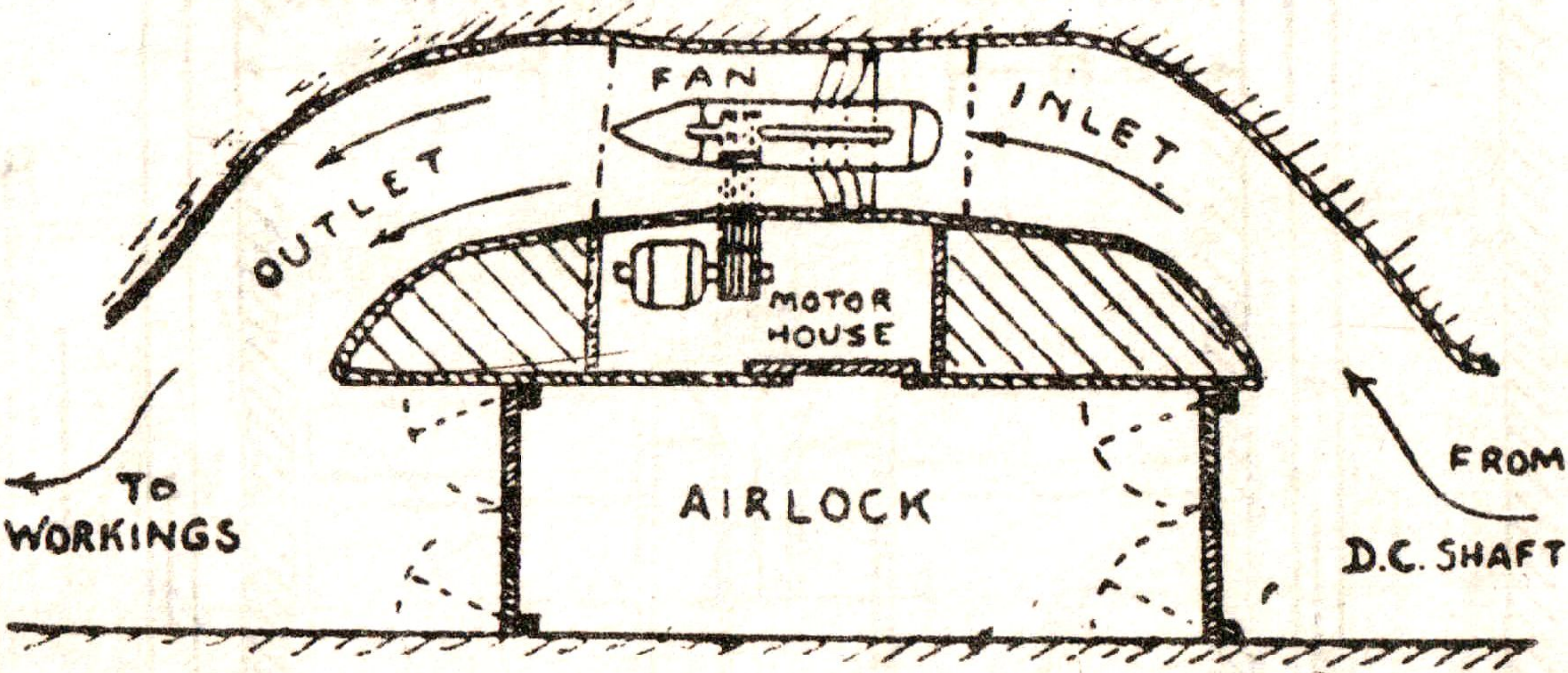


Fig. 3.9. Installation of booster fan; air screw fan.

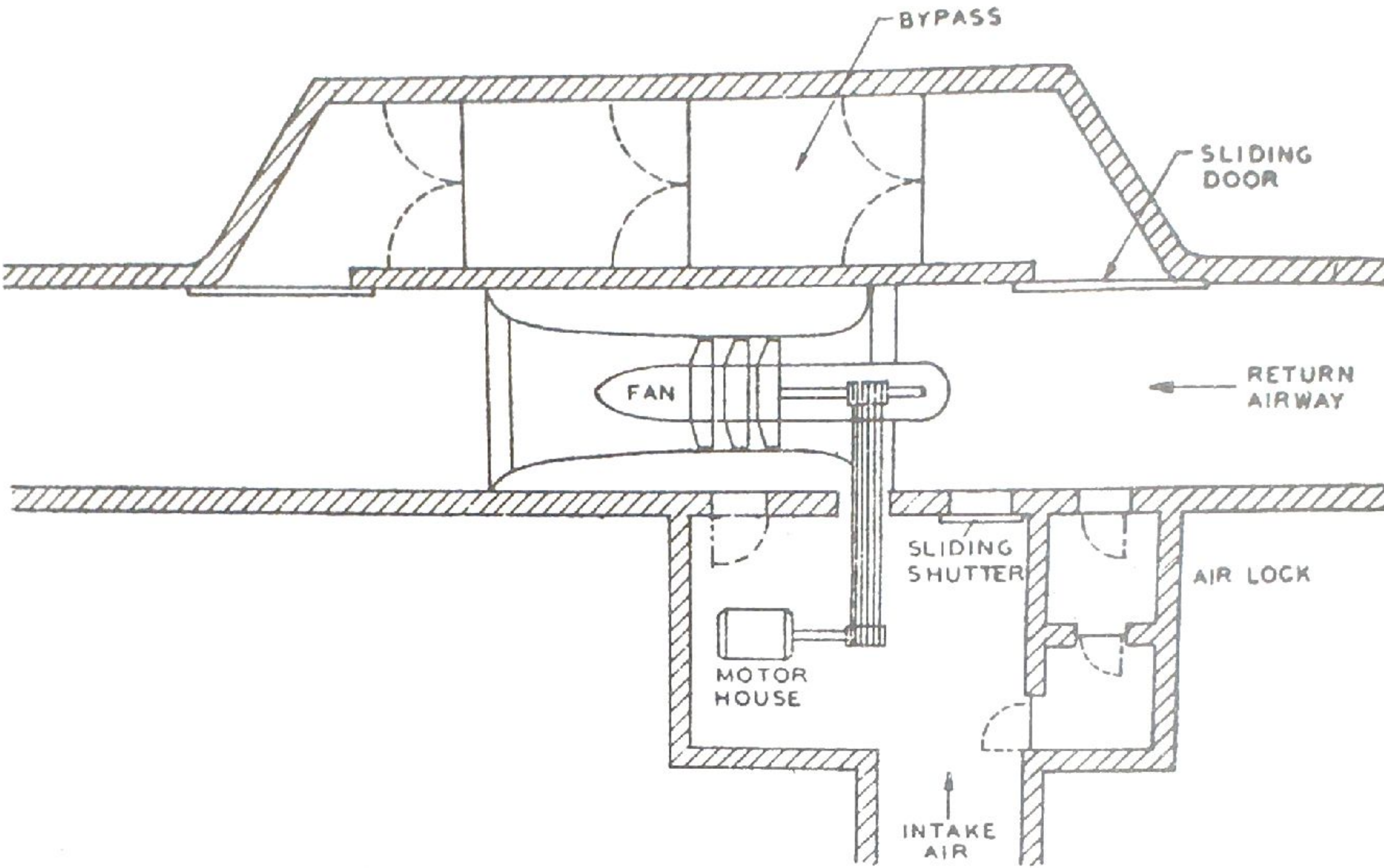


Fig. 6.62 Installation of a 1750mm dia booster fan underground (*after MacFarlane*).

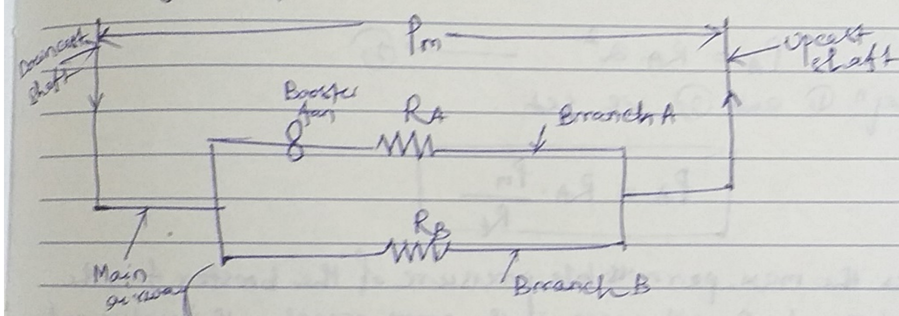
Q: The resistance of two splits A and B are 0.35 Ns^2m^{-8} and 0.05 Ns^2m^{-8} respectively. The combined resistance of the shafts and trunk airways is 0.4 Ns^2m^{-8}. A booster fan is planned to be installed in split A to increase the

quantity flowing through it. Assuming that the surface fan continues to operate at a constant pressure of 1000 Pa, the critical pressure of the booster fan in Pa is ___ ?

Theory: From book Environmental Engg. in Mines by Vutukuri & La

Booster fans:

Sometimes it is necessary to install a booster fan in a branch which may be in parallel with another branch (Fig)



P_m = press. developed by the main fan.

This will increase the airflow in branch A but will reduce the air in the other branch B, because a larger vol. of air will be carried the shafts and main airways, causing an increased pressure drop, there and leaving a smaller portion of the surface fan pressure for

MEMO / TO DO

airflow in branch B.

JANUARY 2009						
S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

The critical condition will be obtained when the airflow in branch B just ceases (or reverses). The entire press. of the main fan is then absorbed in the shafts and ma airways and all the air passes into branch A, the booster fan supplying the pressure to circulate this air round branch A.

If P_m is the pressure developed by the main fan and P_A is the pressure developed by the booster fan in branch A where R_t and R_A are the resistances of the shafts and main airways and branch A, respectively, then for the critical condition when reversal commences in branch B, in which no air now passes,

$$P_m = R_t Q^2 \quad \text{--- (1)}$$

where Q = vol. of flow rate passing in the shafts and main airways. But all the vol. flow rate now passes into branch A, so that

$$P_A = R_A \cdot Q^2 \quad \text{--- (2)}$$

From eqn. (1) and (2) we get

$$\boxed{P_A = R_A \cdot \frac{P_m}{R_t}}$$

This gives the max. permissible pressure of the booster fan, the flow in branch B will cease if the press reaches this value and will be reversed if it is exceeded. Reversed flow will mean recirculation round the two branches.

Soln

$P_m = 1000$ Pa, $R_t = 0.4\ Ns^2m^{-8}$, $R_A = 0.35\ Ns^2m^{-8}$

$R_B = 0.05\ Ns^2m^{-8}$

$$P_A = R_A \cdot \frac{P_m}{R_t} = 0.35 \times \frac{1000}{0.4}$$

$$= 875\ \text{Pa}.$$

The critical press. of booster fan is 875 Pa. (Ans.)

DECEMBER 2008						
S	M	T	W	T	F	S
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
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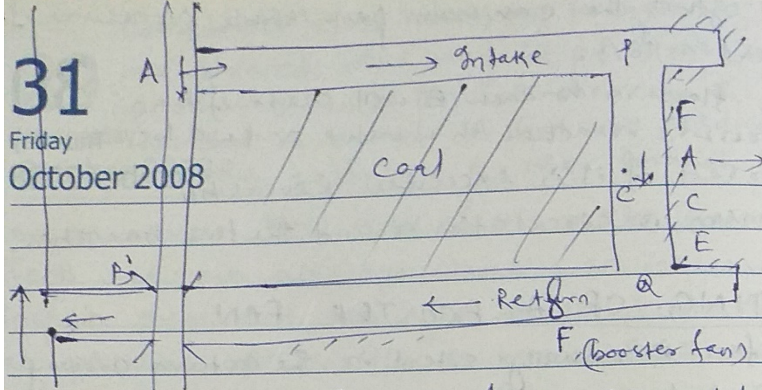
MEMO / TO DO



et us consider a longwall face served by an intake
d return airway separated from each other by mean
f solid coal / or a leaky barrier. Depending on the
nount of leakage, the pressure drop from the entra
the intake to the face and again from the face t
outlet of the return will be in the form of as
own in the fig.

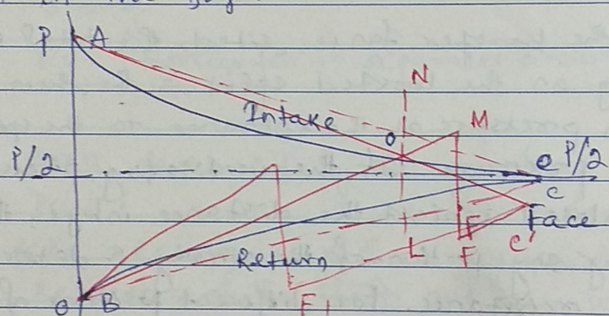


Fig. Pressure gradient along the intake and return airways leading to a face.

However, for the sake of simplicity let us assume a straight line pressure gradient between the entr of the intake or the outlet of the return and the face as shown in red coloured dotted line. This ~~approximation~~ assumption is approximately true for small leakages.

A mine fan generates pressure of 500 Pa which is sufficient to circulate 25 m^3 of air per second through the mine which consists of two splits, A and B, A circulating 15 $m^3 s^{-1}$ and B, 10 $m^3 s^{-1}$. It is desired to increase the quantity in split B to 15 $m^3 s^{-1}$ by installing a booster in it. Calculate the size of the booster if the resistance of the shafts and trunk airways is 0.2 $Ns^2 m^{-8}$.

Let R_A = resistance of split A,
R_B = resistance of split B,
R_T = resistance of shafts and trunk airways,
P_B = pressure generated by the booster fan,
Q_A = quantity in split A after installation of the booster,
Q_B = quantity in split B after installation of the booster
and Q_T = total quantity flowing through the mine after installation of the booster.

Before installation of the booster,

$$500 = R_T \times 25^2 + R_A \times 15^2 = R_T \times 25^2 + R_B \times 10^2$$

or, $R_A = 1.67\ Ns^2 m^{-8}$

and $R_B = 3.75\ Ns^2 m^{-8}$

since $R_T = 0.2\ Ns^2 m^{-8}$.

After installation of the booster,

$$500 = R_T Q_T^2 + R_A (Q_T - Q_B)^2$$

$$= 0.2\ Q_T^2 + 1.67\ (Q_T - 15)^2 \text{ since } Q_B = 15\ m^3 s^{-1},$$

assuming the fan pressure to remain constant.
Solving the above equation, $Q_T = 29\ m^3 s^{-1}$, neglecting the negative value of Q_T.

Now considering flow in split B,

$$500 + P_B = R_B Q_B^2 + R_T Q_T^2$$

$$= 3.75 \times 225 + 0.2 \times 29^2$$

or, $P_B = 512$ Pa.